

Lec 2-2 (Wed, Sep 10, 2025)

- derivative functions (3.2)
 - Differentiability vs continuity (3.2)
 - f' vs Δx and Δy
 - Non-differentiable cases
- Differentiation rules (3.3)
 - Linearity and power rules
 - Product and quotient rules
- Higher-order derivative (3.3)
- Trigonometric functions (3.5)
- Linear motion and cost/production (3.4)

Derivative Functions

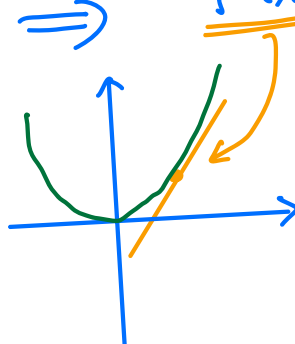
Note that f' itself defines a function: if $f: D \rightarrow \mathbb{R}$, then

$$f'(x) \triangleq \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad \forall x \in D$$

with domain of f' being the set of all $x \in D$ at which f is differentiable.

e.g. $f(x) = x^2$,
 $D = \mathbb{R}$

$\Rightarrow$


$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x, \quad x \in \mathbb{R} \end{aligned}$$

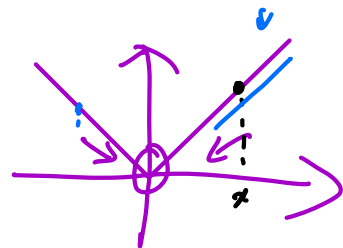
e.g. $f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$,
 $D = \mathbb{R}$

$\Rightarrow$

$$f'(x) = \begin{cases} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{x+h-x}{h} = 1, & x > 0; \\ \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{-(x+h)-(-x)}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1, & x < 0, \end{cases}$$

$f'(0)$ does not exist (---). The domain of f' is $\mathbb{R} \setminus \{0\}$,
i.e., $f'(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$ $(-\infty, 0) \cup (0, \infty)$

[$f'(0)$ does not exist]



• Alternative notation for f'

If $y = f(x)$, then

• f' is the same as y' , $\frac{dy}{dx}$, $\frac{df}{dx}$, $\frac{d}{dx} f$

• $f'(x_0)$ is the same as $y'(x_0)$, $\frac{dy}{dx} \Big|_{x=x_0}$, $\frac{df}{dx} \Big|_{x=x_0}$, $\frac{d}{dx} f(x_0)$.

• Δy is the change of y -value if x -value is changed by Δx :

$$\Delta y \triangleq f(x_0 + \Delta x) - f(x_0)$$

$$\text{Then } \frac{dy}{dx} \Big|_{x=x_0} = f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

• Differentiability implies continuity $\Leftarrow$
 $\Rightarrow$

Thm 3.2.1. If f is differentiable at c , then f is continuous at c .

Pf: • It suffices to show that $\lim_{x \rightarrow c} f(x) = f(c)$.

• Note that

$$\begin{aligned} \left(\lim_{x \rightarrow c} f(x) \right) - f(c) &= \lim_{x \rightarrow c} (f(x) - f(c)) \\ &= \lim_{x \rightarrow c} \left(\frac{f(x) - f(c)}{x - c} \cdot (x - c) \right) \end{aligned}$$

limit law:
product $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \cdot \lim_{x \rightarrow c} (x - c)$

$$= f'(c) \cdot 0$$

$$= 0.$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c).$$

□

• Note that the converse is false.

$$f \text{ cont' at } c \not\Rightarrow f \text{ diff' at } c.$$

See eg 7 and 8 :

$$f(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

not cont' at 0.

$$\lim_{x \rightarrow 0} f(x) = f(0) \quad \Leftarrow \quad f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

cont' at 0 but not diff' at 0.

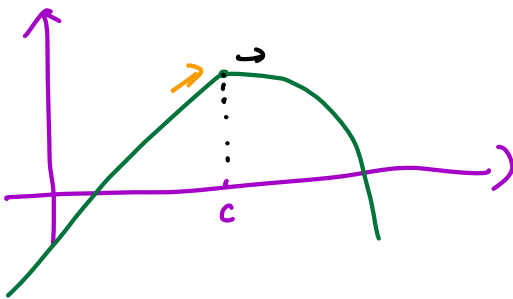
$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$$

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

diff' at 0.

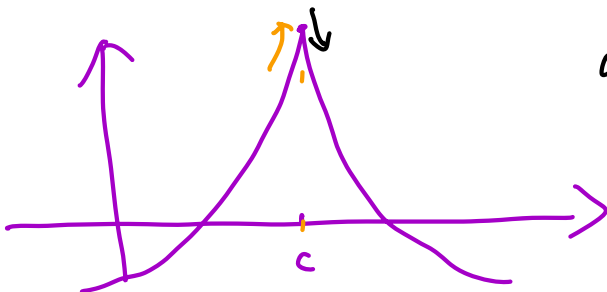
• Not differentiable at a point

①

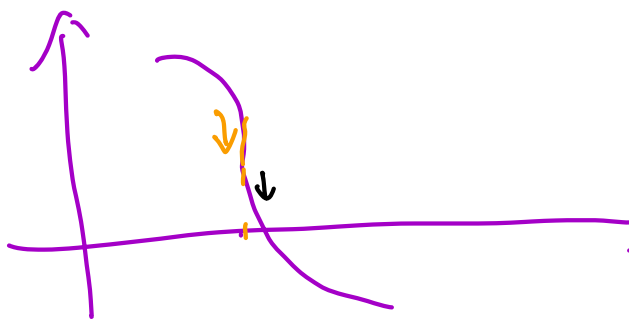


$$f'_-(c) \neq f'_+(c).$$

②



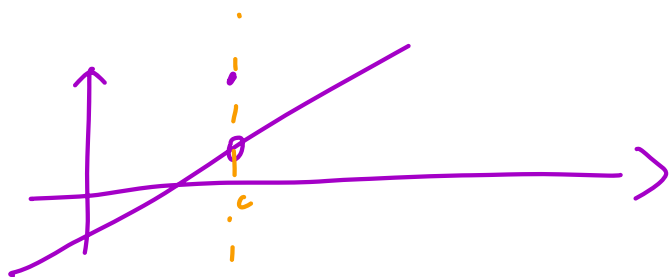
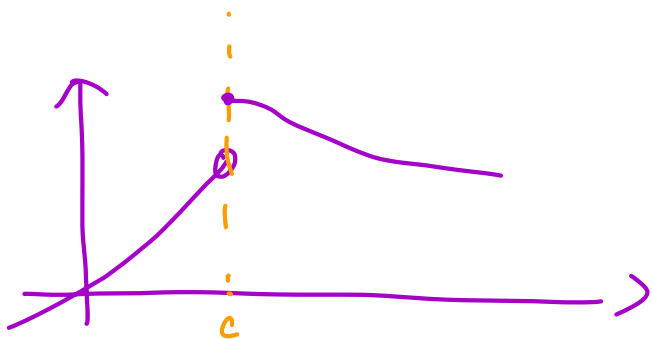
$f'_-(c)$ does not exist as a finite number (∞)
and $f'_+(c)$ does not exist as a finite number ($-\infty$).



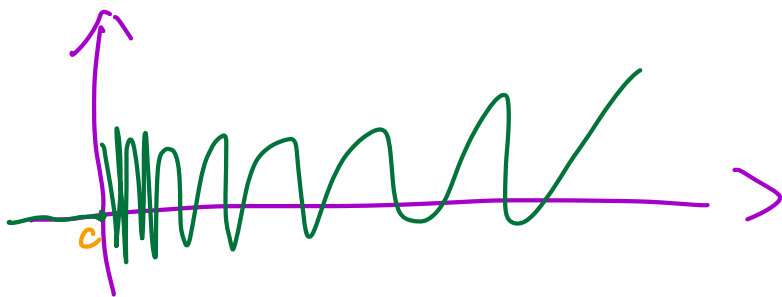
$$\lim_{x \rightarrow c} f'(x) = -\infty$$

i.e., $f'_-(c)$ does not exist as a finite number ($-\infty$)
 and $f'_+(c)$ does not exist as a finite number ($-\infty$).

③



f is not cont' at c .



There are three statements

a) The limit of f at c exists and is finite.

b) f is continuous at c

c) f is differentiable at c

We have : $c) \xRightarrow{\text{implies}} b) \xRightarrow{\text{implies}} a)$

Explanation : a) means

$\lim_{x \rightarrow c} f(x)$ exists and it is finite

[or equivalently, $\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = L$ is finite].

"there is a trend of f at c " .

b) means $\lim_{x \rightarrow c} f(x) = f(c)$.

[or equivalently, $\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = f(c)$].

there is a trend of f at c and the trend is equal to $f(c)$.

c) means $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ exists and is finite .

[or equivalently, $\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} = k$ is finite]

there is a trend of $\frac{f(x) - f(c)}{x - c}$ at c .

If b) holds, we have $\lim_{x \rightarrow c} f(x) = f(c)$

Then $L = f(c)$

Then $\lim_{x \rightarrow c} f(x)$ exists and is finite.

Then a) holds.

If c) holds, we have c belongs to the domain.

And from the above theorem, $\lim_{x \rightarrow c} f(x) = f(c)$

$\Rightarrow$ b) holds

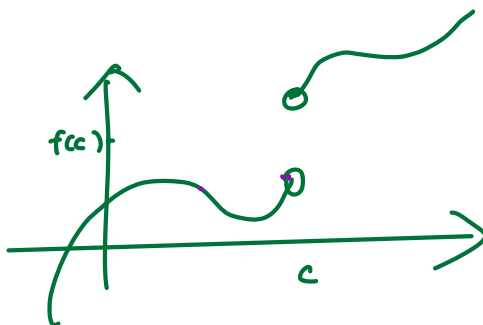
Intuition I: $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = k$

Intuition II: If not continuous,

we have at least a one-side

derivative $= \infty$ or $-\infty$.

$$\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} = \left(\frac{\square^{\neq 0}}{0} \right) = \infty \text{ or } -\infty$$



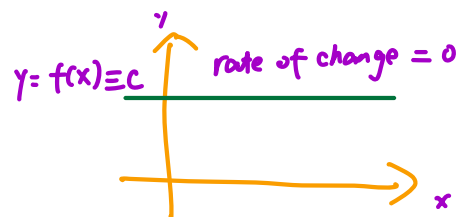
Differentiability $\Rightarrow$ Continuity $\Rightarrow$ finite limit at c .
at c at c

• Differentiation rules (we assume f, g is differentiable below)

① Constant function:

If $f(x) \equiv c$ is a constant function, then

$$f'(x) = 0, \forall x \in \mathbb{R}$$



Pf: $\forall x \in \mathbb{R}, f'(x) = \lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x} = \lim_{y \rightarrow x} \frac{c - c}{y - x} = \lim_{y \rightarrow x} 0 = 0$ □

② Linearity

For any constants $\alpha, \beta \in \mathbb{R}$,

$$[\alpha f(x) + \beta g(x)]' = \alpha \cdot f'(x) + \beta \cdot g'(x), \quad \forall x \in D.$$

In particular,

$$[f(x) + g(x)]' = f'(x) + g'(x).$$

$$[f(x) - g(x)]' = f'(x) - g'(x)$$

$$[\alpha f(x)]' = \alpha \cdot f'(x).$$

[See Pg 134 - 135 in the textbook for the proof for interested readers.]

③ Power rule

If $f(x) = x^\alpha$, where $\alpha \in \mathbb{R}$ is a constant, then

$$f'(x) = \begin{cases} \alpha \cdot x^{\alpha-1}, & \alpha \neq 0 \\ 0, & \alpha = 0 \end{cases}$$

$f(x) = x^0 \equiv \text{const}$
for all x where $\underline{x^\alpha}$ and $x^{\alpha-1}$ are defined.

$$\text{e.g. } \frac{d}{dx} \left[\pi x^3 - \frac{2}{3} x^{\sqrt{2}} + 5x^{\frac{1}{2}} + 48 - x^{-\frac{2}{3}} \right]$$

$$= \pi \cdot 3x^2 - \frac{2}{3} \sqrt{2} x^{\sqrt{2}-1} + 5 \cdot \frac{1}{2} x^{-\frac{1}{2}} - \left(-\frac{2}{3}\right) x^{-\frac{5}{3}}.$$

Pf: step 1. Let us first check the case where $\alpha = n$ is a nonnegative integer.

$$f'(x) = \lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x} = \lim_{y \rightarrow x} \frac{y^n - x^n}{y - x}.$$

$$\text{If } n=1, \quad f'(x) = 1.$$

$$\text{If } n \geq 2, \quad y^n - x^n = (y-x)(y^{n-1} + y^{n-2} \cdot x + y^{n-3} \cdot x^2 + \dots + yx^{n-2} + x^{n-1})$$

RHS = ~~$y^n + y^{n-1} \cdot x + y^{n-2} \cdot x^2 + \dots + y^2 x^{n-2} + yx^{n-1}$~~

$$\begin{aligned}
 f'(x) &= \lim_{y \rightarrow x} \frac{y^n - x^n}{y - x} = \lim_{y \rightarrow x} (y^{n-1} + y^{n-2} \cdot x + y^{n-3} \cdot x^2 + \dots + yx^{n-2} + x^{n-1}) \\
 &= \lim_{y \rightarrow x} y^{n-1} + \dots + \lim_{y \rightarrow x} x^{n-1} \\
 &= x^{n-1} + x^{n-2} \cdot x + \dots + x^{n-1} \\
 &= n \cdot x^{n-1}.
 \end{aligned}$$

Step 2. For $\alpha \in \mathbb{R}$, $\alpha < 0$, we will prove the rule after the quotient rule.

$$f(x) = x^\alpha = x^{-n} = \frac{1}{x^n}.$$

Step 3. For other values of $\alpha \in \mathbb{R}$, we will prove the rule in Chapter 7.

④ product rule.

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x).$$

$$\begin{aligned}
 \text{Pf: } [f(x)g(x)]' &= \lim_{y \rightarrow x} \frac{f(y)g(y) - f(x)g(x)}{y - x} \\
 &= \lim_{y \rightarrow x} \frac{f(y)g(y) - f(y)g(x) + f(y)g(x) - f(x)g(x)}{y - x} \\
 &= \lim_{y \rightarrow x} \frac{f(y)g(y) - f(y)g(x)}{y - x} + \lim_{y \rightarrow x} \frac{f(y)g(x) - f(x)g(x)}{y - x} \\
 &= \lim_{y \rightarrow x} \frac{g(y) - g(x)}{y - x} \cdot f(y) + \lim_{y \rightarrow x} \left[\frac{f(y) - f(x)}{y - x} \cdot g(x) \right]
 \end{aligned}$$

limit law:
= product

$$\lim_{y \rightarrow x} \frac{g(y) - g(x)}{y - x} \cdot \lim_{y \rightarrow x} f(y) + \left(\lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x} \right) \cdot g(x)$$

g, f diff
= f cont

$$g'(x) \cdot f(x) + f'(x) g(x)$$

□.

e.g.

$$y = (x^2 + 1)(x^3 + 1) = x^5 + x^3 + x^2 + 1$$

$$\begin{aligned} y' &= 2x(x^3 + 1) + (x^2 + 1)(3x^2) \\ &= 2x^4 + 2x + 3x^4 + 3x^2 \\ &= 5x^4 + 3x^2 + 2x. \quad \square \end{aligned}$$

⑤ Quotient rule. Assume $g(x) \neq 0$

$$\left[\frac{f(x)}{g(x)} \right]' = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2}$$

Pf:

$$\left[f(x) \right]' = \left[\frac{f(x)}{g(x)} \cdot g(x) \right]' = \left[\frac{f(x)}{g(x)} \right]' g(x) + \frac{f(x)}{g(x)} \cdot g'(x)$$

$$\begin{aligned} \therefore \left[\frac{f(x)}{g(x)} \right]' &= \frac{\left[f(x) \right]' - \frac{f(x)}{g(x)} \cdot g'(x)}{g(x)} \\ &= \frac{g(x) \cdot f'(x) - f(x) g'(x)}{g(x)^2} \end{aligned}$$

□.

[See Chap 3.3 for other proof of quotient rule.]

e.g. $y = \frac{t^2-1}{t^3+1}.$

$$y' = \frac{2t(t^3+1) - (t^2-1) \cdot 3t^2}{(t^3+1)^2}$$

$$= \frac{-t^4 + 3t^2 + 2t}{(t^3+1)^2}.$$

• Recall the derivative of x^α .

Step 2. For $\alpha \in \mathbb{Z}$, $\alpha < 0$, write $\alpha = -n$, where $n \in \{1, 2, \dots\}$.

$$f(x) = x^{-n}.$$

$$\begin{aligned} \text{Then } f'(x) &= (x^{-n})' = \left(\frac{1}{x^n}\right)' \\ &= \frac{0 \cdot x^n - 1 \cdot n x^{n-1}}{x^{2n}} \\ &= - \frac{n x^{n-1}}{x^{2n}} \\ &= -n x^{-n-1} \\ &= \alpha x^{\alpha-1}. \end{aligned}$$

• Higher-order Derivatives.

• $f^{(2)} \triangleq (f')' = f''$: second derivative of f

• In general, for $n \in \mathbb{Z}$, $n \geq 0$:

— $f^{(0)} \triangleq f$, $f^{(1)} \triangleq f'$, $f^{(2)} \triangleq f''$, ..

— $f^{(n)} \triangleq (f^{(n-1)})'$, if $n \geq 1$. : n -th derivative of f .

• Alternative notations : If $y = f(x)$,

— $f''(x) = y'' = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2}{dx^2} y$

— $f^{(n)}(x) = y^{(n)} = \frac{d^n}{dx^n} y$.

e.g. $y = x^3 - 3x^2 + 2$

— $y' = 3x^2 - 6x$

— $y'' = 6x - 6$

— $y''' = 6$

— $y^{(n)} = 0$, $\forall n \geq 4$.

• Derivatives of Trigonometric Functions

$$(\sin x)' = \cos x \quad \forall x \in \mathbb{R}.$$

Pf: $(\sin x)' = \lim_{y \rightarrow x} \frac{\sin y - \sin x}{y - x}$ $\left(\begin{aligned} \sin y &= \sin(x + (y-x)) \\ &= \sin x \cos(y-x) + \cos x \cdot \sin(y-x) \end{aligned} \right)$

$$= \lim_{y \rightarrow x} \frac{\sin x \cos(y-x) + \cos x \cdot \sin(y-x) - \sin x}{y - x}$$

$$= \lim_{y \rightarrow x} \frac{\sin x \cdot [\cos(y-x) - 1]}{y - x} + \lim_{y \rightarrow x} \frac{\cos x \cdot \sin(y-x)}{y - x}$$

$$= \sin x \cdot \lim_{y \rightarrow x} \frac{\cos(y-x) - 1}{y - x} + \cos x \cdot \lim_{y \rightarrow x} \frac{\sin(y-x)}{y - x}$$

$$\stackrel{h=y-x}{=} \sin x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \cos x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h}$$

$$\cos(2\alpha) = \cos^2 \alpha - \sin^2 \alpha$$

$$= \sin x \cdot \lim_{h \rightarrow 0} \frac{\cos(2 \cdot \frac{h}{2}) - 1}{h} + \cos x \cdot 1$$

$$= \sin x \cdot \lim_{h \rightarrow 0} \frac{-2\sin^2(\frac{h}{2})}{h} + \cos x$$

$$= \sin x \cdot (-1) \cdot \lim_{h \rightarrow 0} \frac{\sin(\frac{h}{2})}{\frac{h}{2}} \cdot \lim_{h \rightarrow 0} \sin(\frac{h}{2}) + \cos x$$

$$= \sin x \cdot (-1) \cdot 1 \cdot 0 + \cos x = \cos x. \quad \square$$

$$(\cos x)' = -\sin x \quad \forall x \in \mathbb{R}.$$

Pf: $\lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} = \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h}$

$$= \lim_{h \rightarrow 0} \cos x \frac{\cos h - 1}{h} - \lim_{h \rightarrow 0} \sin x \frac{\sin h}{h}$$

$$= \cos x \cdot 0 - \sin x \cdot 1 = -\sin x. \quad \square$$

$$(\tan x)' = \frac{1}{\cos^2 x}$$

$$(\sec x)' = \left(\frac{1}{\cos x}\right)' = \frac{\tan x}{\cos x}.$$

$$\begin{aligned} \text{Pf: } (\tan x)' &= \left(\frac{\sin x}{\cos x}\right)' = \frac{\cos x \cdot \cos x - \sin x (-\sin x)}{\cos^2 x} \\ &= \frac{1}{\cos^2 x}. \end{aligned}$$

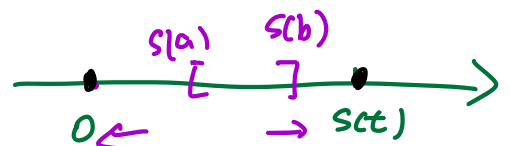
$$\begin{aligned} (\sec x)' &= \left(\frac{1}{\cos x}\right)' = \frac{0 - 1 \cdot (-\sin x)}{\cos^2 x} \\ &= \frac{\sin x}{\cos^2 x} = \frac{\tan x}{\cos x}. \end{aligned}$$

□.

Application : Motion along a line

suppose an object is moving along a line, where

its position at time t is $s = s(t)$



Note that s can be negative.

Def: The displacement of the object over a time interval $[a, b]$ is $s(b) - s(a)$.

- The Velocity of the object at time t is

$$v(t) \triangleq \frac{ds}{dt} = \lim_{\Delta t \rightarrow 0} \frac{s(t+\Delta t) - s(t)}{\Delta t}$$

- The speed of the object at time t is

$$|v(t)|$$

- the acceleration of the object at time t is

$$a(t) \triangleq \frac{dv}{dt} = \frac{d^2}{dt^2} s$$

- The jerk of the object at time t is

$$j(t) \triangleq \frac{da}{dt} = \frac{d^2}{dt^2} v = \frac{d^3}{dt^3} s$$

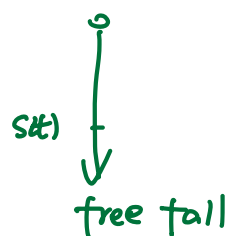
e.g.

$$s(t) = \frac{1}{2} \cdot 9.8 \cdot t^2$$

$$v(t) = \frac{1}{2} \cdot 9.8 \cdot 2t = 9.8t$$

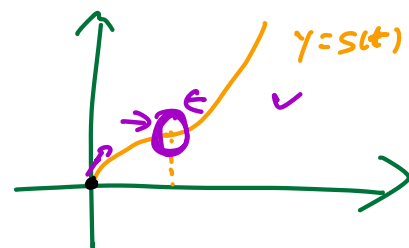
$$a(t) \equiv 9.8$$

$$j(t) \equiv 0$$



e.g.

$$s(t) = \begin{cases} 1 - (t-1)^2, & 0 \leq t \leq 1 \\ (t-1)^2 + 1, & t > 1 \end{cases}$$



$$v(t) = \begin{cases} -2(t-1) & , 0 \leq t < 1 \\ 2(t-1) & , t > 1 \end{cases} \quad (t=1)$$

$$\text{Compute: } s'_+(1) = \lim_{t \rightarrow 1+} \frac{s(t) - s(1)}{t - 1} = \lim_{t \rightarrow 1+} \frac{(t-1)^2 + 1 - 1}{t - 1} = \lim_{t \rightarrow 1+} (t-1) = 0$$

$$s'_-(1) = \lim_{t \rightarrow 1-} \frac{s(t) - s(1)}{t - 1} = \lim_{t \rightarrow 1-} \frac{1 - (t-1)^2 - 1}{t - 1} = \lim_{t \rightarrow 1-} -(t-1) = 0$$

$$\therefore s_+'(1) = s_-'(1) = 0 \Rightarrow V(1) = 0.$$

$$\therefore V(t) = \begin{cases} -2(t-1) & , \quad 0 \leq t < 1. \\ 0 & , \quad t = 1. \\ 2(t-1) & , \quad t > 1. \end{cases}$$

Next, we compute

$$a(t) = \begin{cases} V'(t) = -2 & , \quad 0 \leq t < 1; \\ V'(t) = 2 & , \quad t > 1; \end{cases}$$

$$\therefore V_+'(1) = \lim_{t \rightarrow 1^+} \frac{V(t) - V(1)}{t - 1} = \lim_{t \rightarrow 1^+} \frac{2(t-1) - 0}{t - 1} = \lim_{t \rightarrow 1^+} 2 = 2.$$

$$V_-'(1) = \lim_{t \rightarrow 1^-} \frac{V(t) - V(1)}{t - 1} = \lim_{t \rightarrow 1^-} \frac{-2(t-1) - 0}{t - 1} = \lim_{t \rightarrow 1^-} (-2) = -2.$$

$\therefore V'(1)$ (or $a(1)$) does not exist.

$$\therefore a(t) = \begin{cases} -2 & , \quad 0 \leq t < 1; \\ 2 & , \quad t > 1; \\ \text{not defined} & , \quad t = 1. \end{cases}$$

□.